e-Commerce Store Dash Board using Excel

This assignment demonstrates the complete data analytics workflow using Microsoft Excel, starting from raw data import to final insights and visualization. The dataset used represents e-commerce sales transactions four different sheets Customer_Dim, Product_Dim, Store_Dim & Sales_fact. It contains valuable information's like Sales_ID, Order_Date, Customer_ID, Product_ID, Store_ID, Quantity, Unit_Price, Discount, Payment_Type, Total_Amount, Name_New, Age, Gender, State, Loyalty_Level, Brand, Product_Name & Store_Type.

- Step1: First reviewed all four sheets of data set & identified the typo error, Blank space, irregular text format. Then rectified it.

- ✓ In Customer_Dim sheet,
  - Customer_ID column has type error. Rectified it by Find & replace method.
  - Name Column has irregular text format. Rectified it by using Trim and proper Funtions.
  - Loyalty_Level column had blank cells. Rectified it by using IF(ISBLANK(K2),"Nil",K2) Function to find the blank cell and replace it by "Nil" value.

- ✓ In Product_Dim sheet,
  - Product_ID column has type error. Rectified it by Find & replace method.
  - Cost & Stock column had blank cells. Rectified it by using IF(ISBLANK(G2),AVERAGE($G$2:$G$101),G2) Function to find the blank cell and replace it by "AVERAGE" value.

- ✓ In Sales_Fact sheet,
  - Created a new column for Month, Extracted the month from the Order_Date column by using the =TEXT(B130,"mmm") function.
  - Quantity Column Level column had blank cells. Rectified it by using IF(ISBLANK(G130),"0", G130) Function to find the blank cell and replace it by "Zero" value.
  - Unit_Price & Tota_Amount column had blank cells. Rectified it by using IF(ISBLANK(M130),AVERAGE(M130:M2129),M130) Function to find the blank cell and replace it by "AVERAGE" value.

Step 2:

- In the Sales_Fact sheet created new columns from following category's - Name_New, Age, Gender, State, Loyalty_Level, Brand, Product_Name & Store_Type.

- Then keeping Customer_Id has the reference value, used VLOOKUP function to pull the data for Name_New, Age, Gender, State & Loyalty_Level from the Customer_Dim sheet. VLOOKUP(D130,Customer_Dim!B$1:L$501,4,0)

- Using Product_ID _Id has the reference value, applying VLOOKUP function to pull the data for Brand & Product_Name from the Product_Dim sheet. VLOOKUP(E130,Product_Dim!$A$1:$I$101,5,0)

- Likewise used Store_ID has reference pulled data from Store_Type column from Store_Dim sheet using VLOOK Function. VLOOKUP(F130,Store_Dim!$A$1:$E$21,5,0)

Step 3:

Created a new sheet, named it has FINAL_ALL_CUS_PRO_STORE_SALES & Consolidated all the data for applying pivot methods by comparing specific datas.

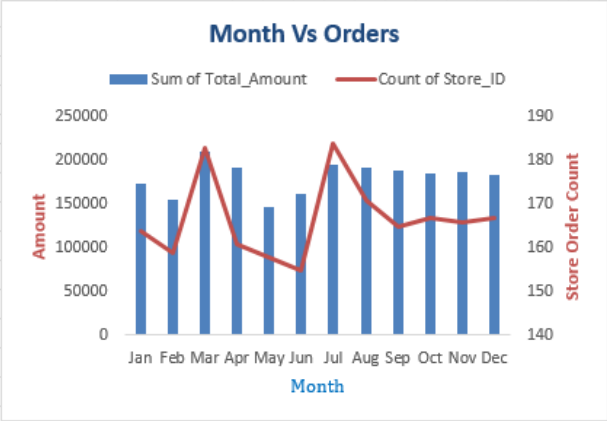
Step 4:

- Selected all the data range in the FINAL_ALL_CUS_PRO_STORE_SALES sheet,
- Clicked on Insert tab
- Clicked pivot table
- In create pivot table , selected new worksheet option.
- In the pivot table sheet, renamed it pivot area.

- Step 5: Creating Pivot Table and charts.

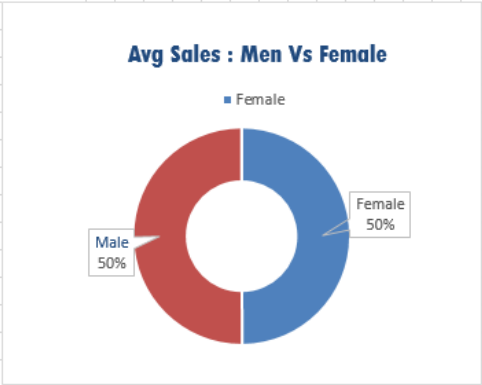
Created Month Vs Order cluster colum, by using Month, Total_Amount (sum) & Store_ID (count). Used Combo option to get secondary axis.

Row Labels	Sum of Total_A	Count of Store_ID
Jan	174449.14	164
Feb	155073.9391	159
Mar	210704.522	183
Apr	192864.0353	161
May	146511.9629	158
Jun	162640.0165	155
Jul	195601.0806	184
Aug	192164.3513	171
Sep	188232.3108	165
Oct	186168.2067	167
Nov	187386.1682	166
Dec	183534.3284	167
Grand Total	**2175330.062**	**2000**



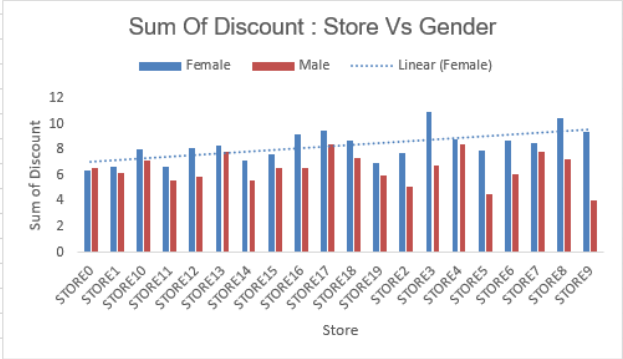
Created a Donut pie chart, by creating pivot for the values of gender and total amount (Avg) to showcase avg sales between men and female.

Row Labels	Average of Total_Amount
Female	$1,085.98
Male	$1,089.71
Grand Total	**$1,087.67**



Created a cluster column chart with linear line to show the trend by using the store_id, gender & discount (sum)

Sum of Discount	Column Lab		
Row Labels	Female	Male	Grand Total
STORE0	6.39	6.57	12.96
STORE1	6.63	6.21	12.84
STORE10	8	7.14	15.14
STORE11	6.69	5.63	12.32
STORE12	8.12	5.93	14.05
STORE13	8.35	7.83	16.18
STORE14	7.11	5.63	12.74
STORE15	7.61	6.55	14.16
STORE16	9.21	6.53	15.74
STORE17	9.53	8.4	17.93
STORE18	8.71	7.31	16.02
STORE19	6.96	5.94	12.9
STORE2	7.77	5.15	12.92
STORE3	10.95	6.77	17.72
STORE4	8.77	8.37	17.14
STORE5	7.96	4.57	12.53
STORE6	8.66	6.07	14.73
STORE7	8.55	7.88	16.43
STORE8	10.44	7.21	17.65
STORE9	9.42	4.06	13.48
Grand Total	**165.83**	**129.75**	**295.58**



Created Stock Column chart for total order between state and loyalty holder using data of state, Loyalty & qty ordered.

Count of Quantit	Column Lab					
Row Labels	Bronze	Gold	NIL	Platinum	Silver	Grand Total
Alabama	13			14	19	46
Alaska	6	8		20	7	41
Arizona	4	13		11	13	41
Arkansas	3	2		7	14	26
California	9	14		13	17	53
Colorado	2			6		8
Connecticut	18	9		10	18	55
Delaware	12			5	15	32
Florida	15	7		20	10	52
Georgia	2	15		13	20	50
Hawaii	11			16	5	32
Idaho	2	3		3	3	11
Illinois	13	9		8		30
Indiana		2		25	12	39
Iowa	8	17		14	8	47
Kansas	1	9	2	21	14	47
Kentucky	6	6		13		25
Louisiana	12	10		17		39
Maine	6	13		17		36
Maryland	4	12		12		28
Massachusetts	7				14	21
Michigan	6	16		17	6	45
Minnesota	16			13	11	40

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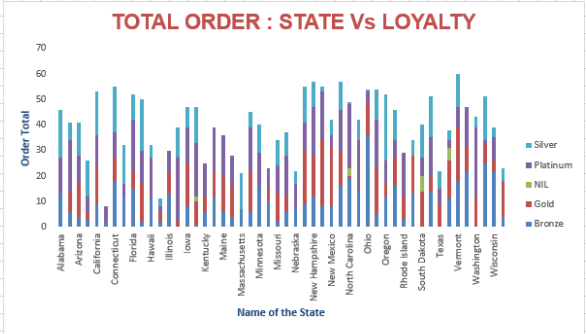
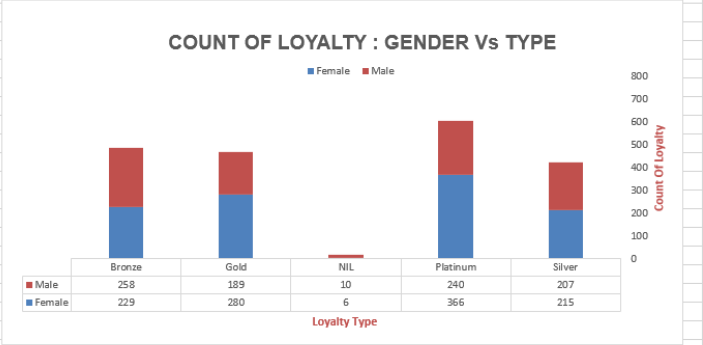


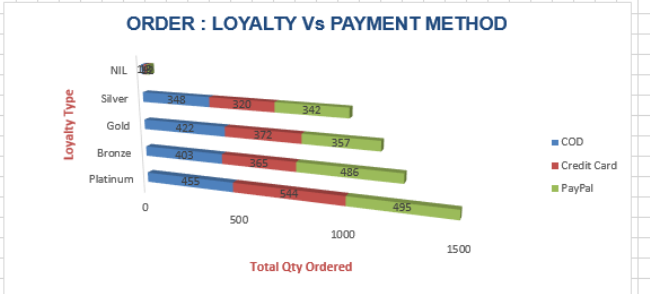
Created a stocked column chart to show case the count of loyalty between gender and loyalty type. Pivoted the information of loyalty type & gender.

Count of Loyalty_ Column Lab			
Row Labels	Female	Male	Grand Total
Bronze	229	258	487
Gold	280	189	469
NIL	6	10	16
Platinum	366	240	606
Silver	215	207	422
Grand Total	1096	904	2000



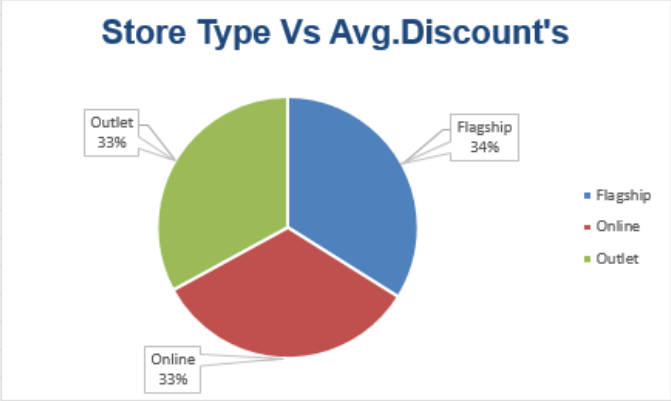
Created a pivot table comparing royalty type with payment method. Used bar chart type to show case the breakups of the order placed.

Sum of Quantity	Column Lab			
Row Labels	COD	Credit Card	PayPal	Grand Total
Platinum	455	544	495	1494
Bronze	403	365	486	1254
Gold	422	372	357	1151
Silver	348	320	342	1010
NIL	14	13	12	39
Grand Total	1642	1614	1692	4948

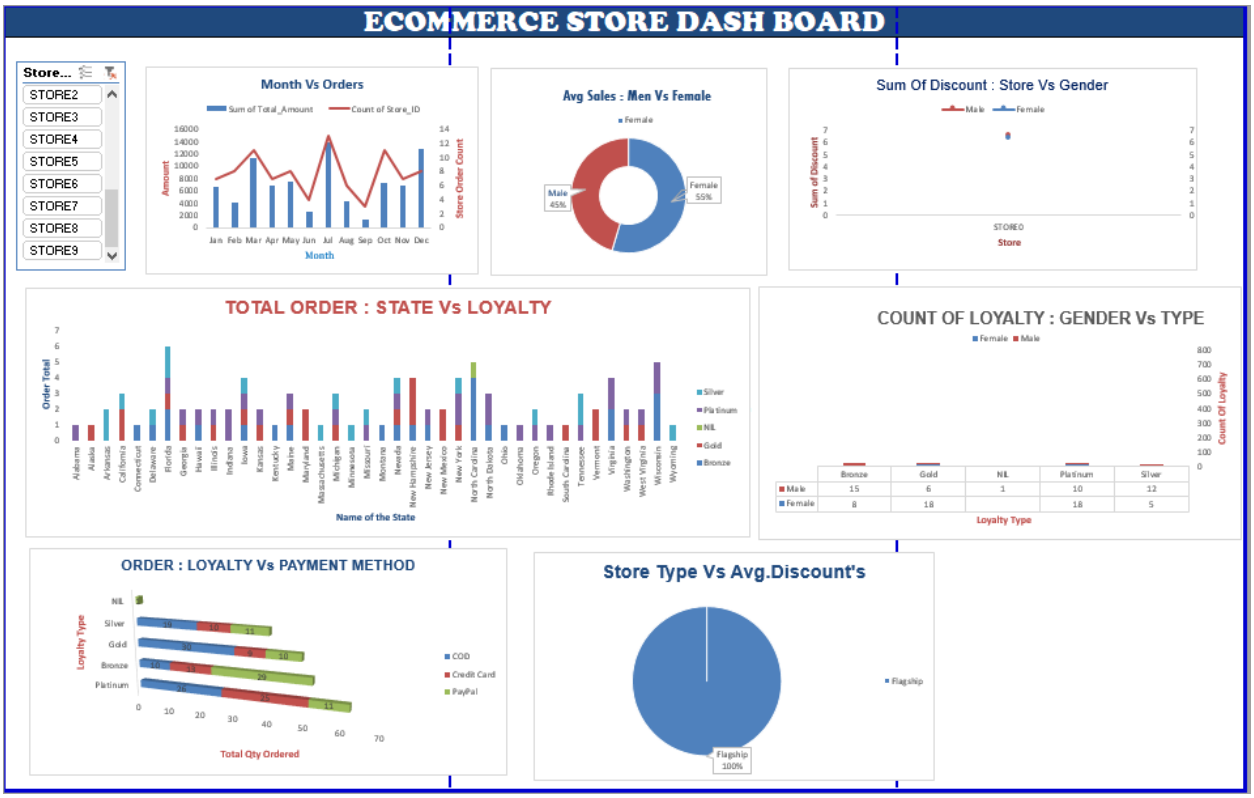


Created a pie chart to show case the Avg discount availed by the customer fro different store types for the order. Pivoted using data of store type and avg of discount offered.

Row Labels	Average of Discount
Flagship	0.150264484
Online	0.146959707
Outlet	0.145979112
Grand Total	0.148086172



- Step 6: Creating the e-commerce dashboard.
  - Created a new sheet.
  - In the view tab, unchecked the grind line to make the sheet blank.
  - Merged the first two rows, entered the dash board name
  - Copy and pasted all the pivot charts & arranged it
  - Selected a chart and clicked on insert, then clicked on slicer option,
    - In the drop down selected store option.
    - Selected the slicer, from option tab, clicked on report connection
    - Select all the category, so that all the chart get connected for interactive dashboard.



Conclusion

The analysis helped identify top-performing store and sales trends, supporting data-driven decision-making. This assignment highlights how Excel can be effectively used as a data analytics tool for cleaning, analyzing, and visualizing business data in a structured manner.

Excel Work sheet Enclosed for referrence:
https://docs.google.com/spreadsheets/d/14r0fct-Fin6An92UuIxt733dvyo0TNpv/edit?usp=drive_link&ouid=106177419796890872629&rtpof=true&sd=true